PART 1: INTRODUCTION TO LOGS

Research and explain the following:

1. **What are Logs?**
2. **Why are logs important in Cyber Security?**
3. **Types of Logs:**
    - **System Logs**
    - **Application Logs**
    - **Web Server Logs**
    - **Authentication Logs**
    - **Firewall Logs**

PART 2: WINDOWS EVENT VIEWER ANALYSIS

Open Event Viewer on Windows.

Go to:

Windows Logs → Security

Analyze and record:

1. **Total number of events.**
2. **Latest login event.**
3. **Latest logout event.**
4. **Failed login attempts (if any).**
5. **Event ID of login events.**

Take screenshots.

PART 3: WEB SERVER LOG ANALYSIS

Download or create a sample Apache/Nginx log file.

Analyze the following:

1. **Client IP Address**
2. **Request Method (GET/POST)**
3. **Requested URL**
4. **Response Code**
5. **User-Agent**

Identify:

- **Most frequently accessed page**
- **Number of unique IP addresses**
- **Number of 404 errors**

PART 4: INCIDENT INVESTIGATION

Assume the following scenario:

"A company reported multiple failed login attempts on its web application."

Tasks:

1. **What information would you collect?**
2. **Which logs would you analyze?**
3. **What indicators suggest a possible attack?**
4. **What immediate actions should be taken?**

PART 5: SUSPICIOUS ACTIVITY DETECTION

Review the collected logs and identify:

- **Multiple failed logins**
- **Unusual login times**
- **Repeated requests from same IP**
- **Access to non-existing pages (404)**
- **Suspicious User-Agents**

Document all findings.

ASSIGNMENT

Prepare a report containing:

1. **Introduction to Log Analysis**
2. **Event Viewer Screenshots**
3. **Web Log Analysis Results**
4. **Incident Investigation Process**
5. **Suspicious Activities Found**
6. **Recommendations**

Create a timeline of events for a security incident:

Example:

10:00 AM - User Login
10:05 AM - Multiple Failed Logins
10:10 AM - Suspicious IP Detected
10:15 AM - Account Locked

Explain how you would respond to each event.

SUBMISSION

- **PDF or Word Document**

INTRODUCTION TO LOG ANALYSIS

What are Logs?

Logs are digital records of events generated by operating systems, applications, web servers, and security devices. These records contain important details such as user activities, login attempts, system errors, and network traffic. Logs help administrators monitor and understand what is happening inside a system.

Logs act as evidence of activities performed within a system and are useful during troubleshooting and forensic investigations.

Why are Logs Important in Cyber Security?

Logs play an important role in cyber security because they help detect suspicious activities and security breaches. They provide information about who accessed the system, when they accessed it, and what actions were performed.

Importance of logs:

- Detect unauthorized access
- Monitor failed login attempts
- Identify malware activities
- Track changes in system files
- Investigate incidents
- Maintain compliance and auditing

Types of Logs

Log Type	Description
System Logs	Record operating system events such as startup, shutdown, and errors
Application Logs	Store events generated by software applications
Web Server Logs	Record HTTP requests, responses, and user activities on websites

Authentication Logs	Record successful and failed login attempts
Firewall Logs	Track incoming and outgoing network traffic

WINDOWS EVENT VIEWER ANALYSIS

Windows Event Viewer is a built-in tool used to monitor and analyze system logs.

Steps Performed:

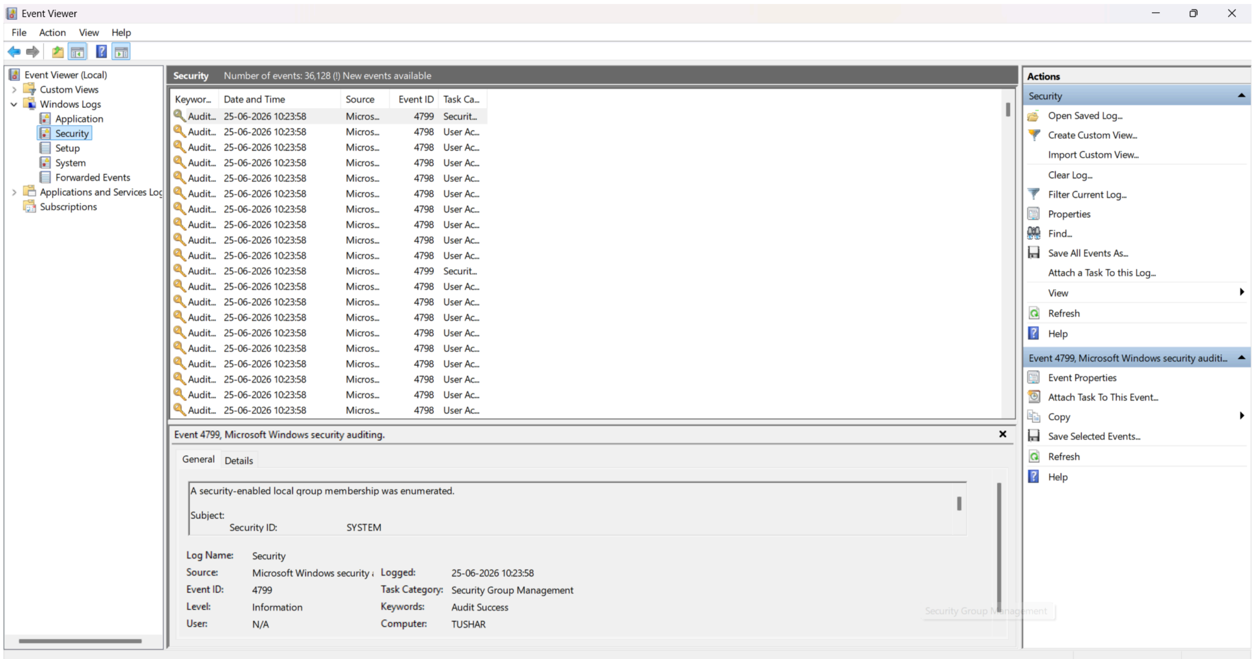
1. Opened Run dialog using **Windows + R**
2. Typed **eventvwr.msc**
3. Opened **Windows Logs**
4. Selected **Security Logs**
5. Filtered logs based on Event IDs:
    - 4624 (Successful Login)
    - 4634 (Logout)
    - 4625 (Failed Login)

Security Log Overview

The security log contains information related to login, logout, account access, and failed authentication attempts.

Total Number of Events Recorded: 36,128

Figure 1: Security Log Overview

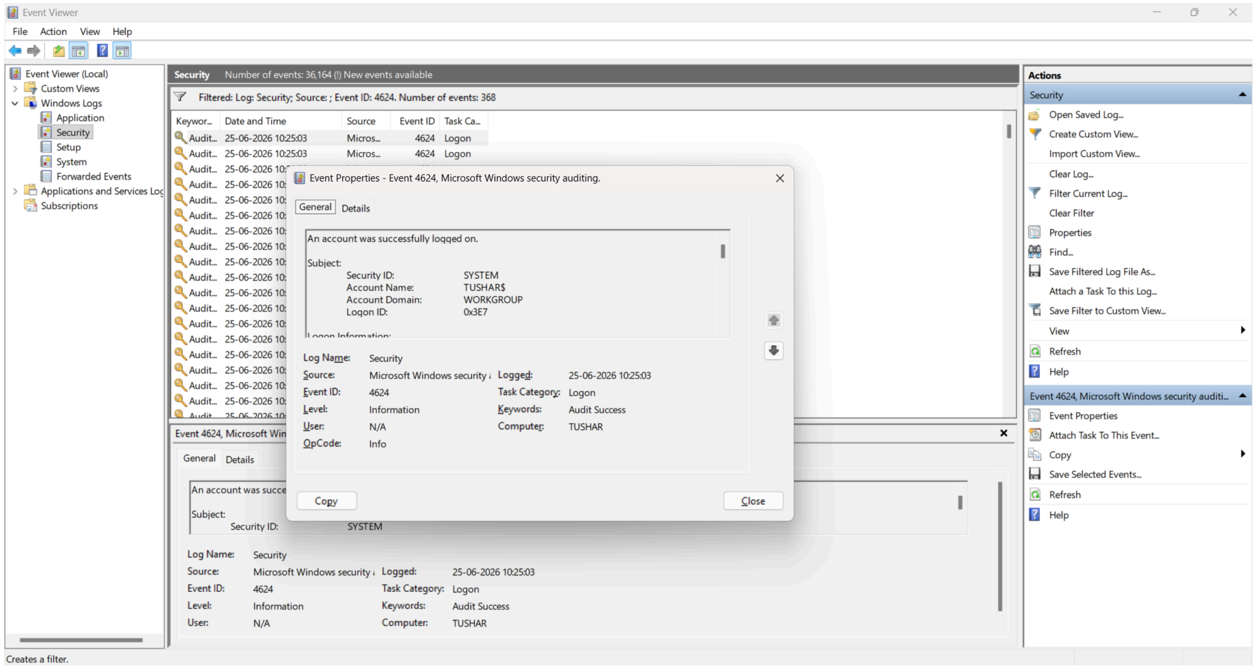


Latest Login Event Analysis

The latest successful login event was found using Event ID 4624.

Field	Value
Event ID	4624
Event Type	Successful Login
Date and Time	25-06-2026 10:25:03
Account Name	TUSHAR$

Figure 2: Latest Login Event

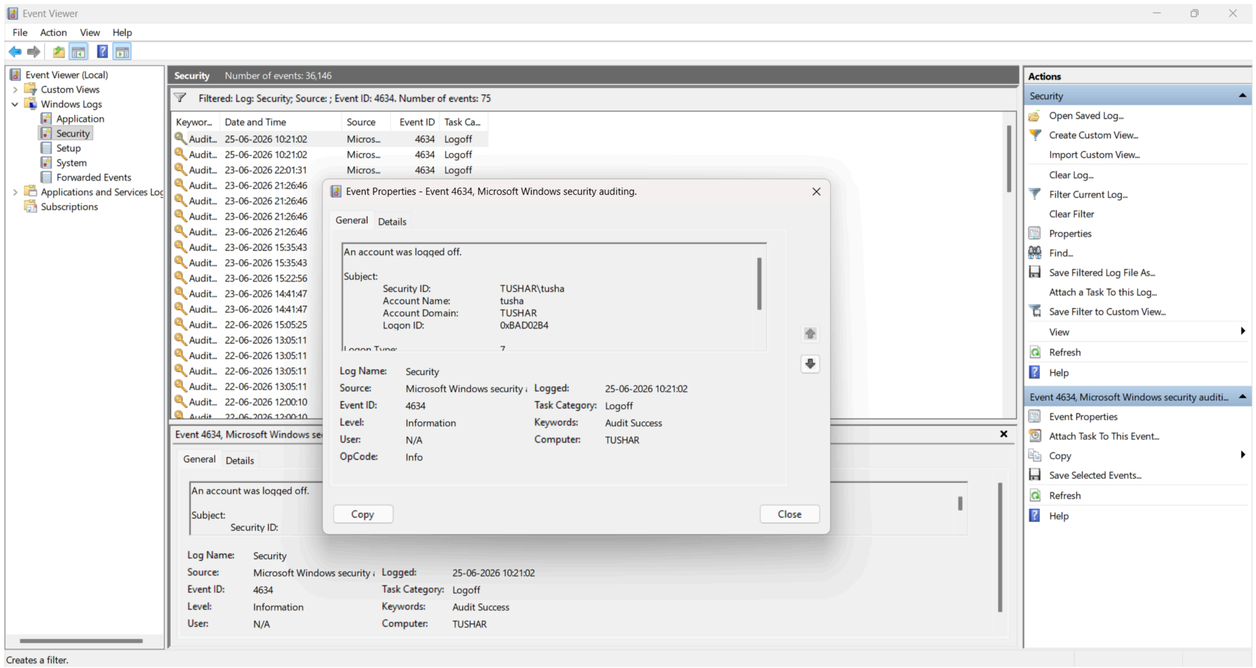


Latest Logout Event Analysis

The latest logout event was found using Event ID 4634.

Field	Value
Event ID	4634
Event Type	Logoff
Date and Time	25-06-2026 10:21:02
Account Name	tusha

Figure 3: Latest Logout Event



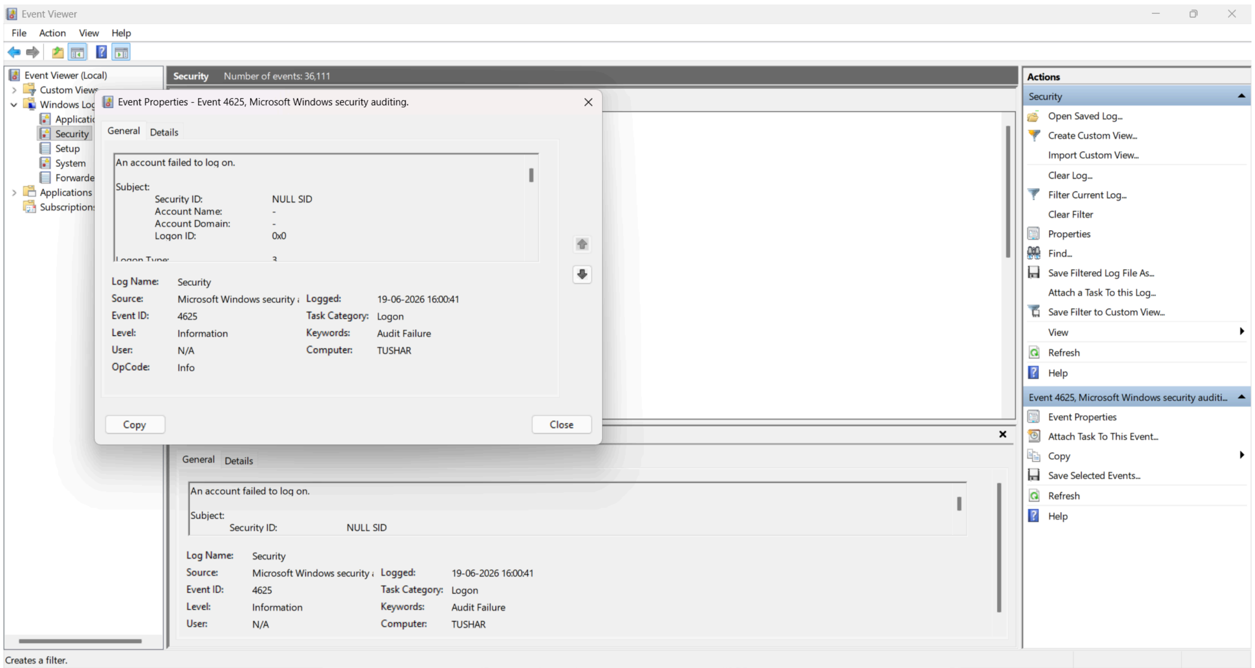
Failed Login Attempt Analysis

Failed login attempts were identified using Event ID 4625.

Field	Value
Event ID	4625
Event Type	Failed Login
Date and Time	19-06-2026 16:00:41
Status	Audit Failure

This indicates that an unauthorized or incorrect login attempt occurred.

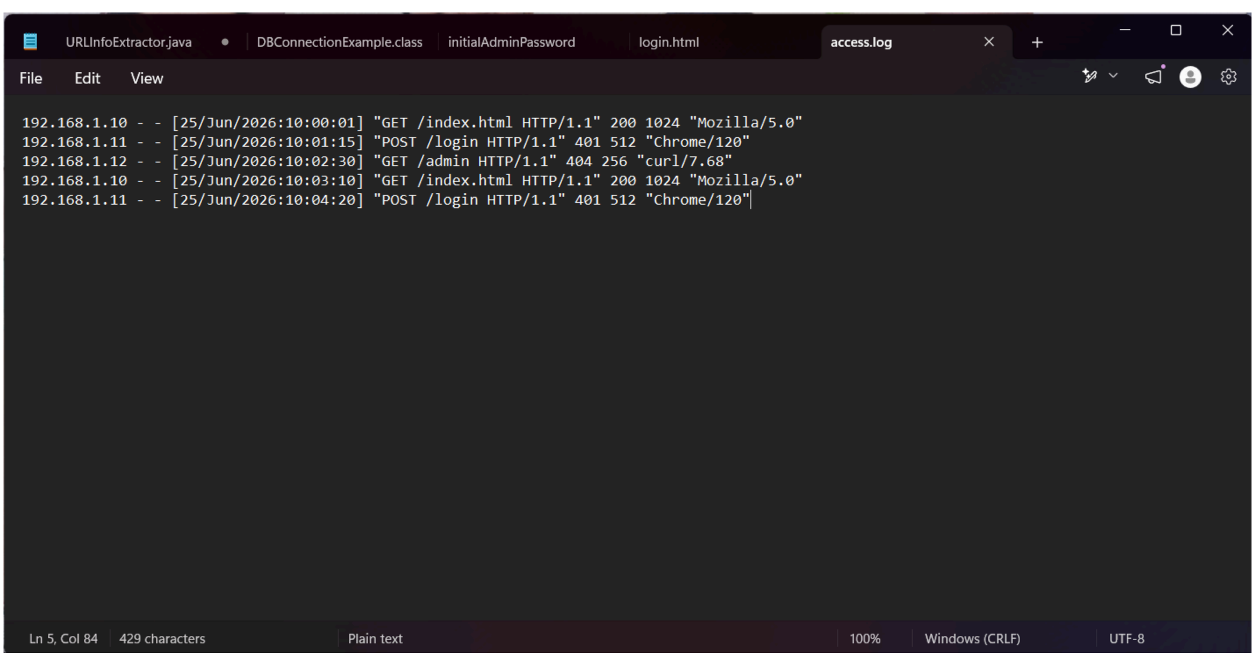
Figure 4: Failed Login Attempt



WEB SERVER LOG ANALYSIS

A sample Apache web server log file was created and analyzed.

Figure 5: Sample Web Server Log File



Log Analysis Table

Client IP Address	Request Method	Requested URL	Response Code	User-Agent
192.168.1.10	GET	/index.html	200	Mozilla/5.0
192.168.1.11	POST	/login	401	Chrome/120
192.168.1.12	GET	/admin	404	curl/7.68
192.168.1.10	GET	/index.html	200	Mozilla/5.0
192.168.1.11	POST	/login	401	Chrome/120

Findings from Web Logs

Most Frequently Accessed Page:

/index.html (Accessed 2 times)

Number of Unique IP Addresses:

3

Number of 404 Errors:

1

INCIDENT INVESTIGATION

Scenario:

A company reported multiple failed login attempts on its web application.

Information to Collect

To investigate this incident, the following information should be collected:

- Source IP addresses
- Target usernames
- Timestamps of login attempts
- Number of failed attempts
- User-Agent details
- Geolocation of IP
- Firewall alerts

Logs to Analyze

The following logs are important:

- Authentication logs
- Web server logs
- Firewall logs
- IDS/IPS logs
- Application logs

Indicators of Possible Attack

The following indicators may suggest an attack:

- Multiple failed login attempts
- Same IP targeting multiple accounts
- Repeated login requests
- Suspicious User-Agent
- Access to restricted pages
- Multiple 404 errors

These indicators may suggest brute-force or reconnaissance attacks.

Immediate Actions

To respond immediately:

1. Block suspicious IP address
2. Lock targeted accounts
3. Reset passwords
4. Enable Multi-Factor Authentication
5. Alert system administrator
6. Monitor logs continuously

SUSPICIOUS ACTIVITY DETECTION

After reviewing collected logs, the following suspicious activities were detected:

Suspicious Activity	Evidence
Multiple failed logins	Event ID 4625
Unusual login attempts	Detected in logs
Repeated requests from same IP	192.168.1.11
Access to non-existing pages	/admin (404)
Suspicious User-Agent	curl/7.68

SECURITY INCIDENT TIMELINE

Time	Event	Response
10:00 AM	User Login	Monitor activity
10:01 AM	Failed Login Attempt	Verify credentials
10:02 AM	Access to /admin page	Investigate source IP
10:03 AM	Repeated homepage access	Normal monitoring
10:04 AM	Another failed login	Possible brute-force detection
10:15 AM	Account Locked	Prevent further attacks

RECOMMENDATIONS

Based on the analysis, the following recommendations are suggested:

- Enable Multi-Factor Authentication (MFA)
- Implement account lockout after multiple failures
- Monitor logs regularly
- Use SIEM tools for automated detection
- Block suspicious IP addresses
- Review failed login patterns
- Monitor unusual User-Agent strings
- Investigate repeated 404 errors

CONCLUSION

Log analysis is an important part of cyber security. It helps in identifying suspicious activities, failed login attempts, and potential attacks. In this practical, Windows Event Viewer and web server logs were analyzed successfully. Several suspicious activities were identified, and preventive measures were suggested.

This practical demonstrates how log analysis can be used for security monitoring and incident investigation.